

# Ian McCallum

## Founder & AI Engineer

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Founder of Beat the Clock and sole architect of Clockwork, an AI operations platform for trades businesses that turns a missed call into a drafted reply, a priced quote, a confirmed appointment, and a collected invoice. I work in Python, prompt engineering, and applied AI systems design, with a bias toward software that refuses to guess when the answer costs somebody money. Studying Finance & Data Science at the University of Illinois' Gies College of Business.

### EXPERIENCE

#### Beat the Clock — Founder & Lead Engineer

Dec 2024 – Present

- Clockwork** — an AI operations platform for trades businesses. Ten automated stages across eight screens; built and demo-ready, with founding pricing open to the first 25 Illinois contractors.
- Designed and built the entire platform as sole architect and engineer, in Python — from catching a missed call and qualifying the lead to pricing the quote, chasing the follow-up, collecting payment, and drafting tomorrow's schedule overnight.
  - Built the product around an approval-gated AI agent:** every message is drafted for the owner to approve, and the agent only earns permission to send a message type on its own after enough drafts pass unedited — a deliberate two-step switch, revocable in one tap.
  - Separated deterministic pricing from model output** so quotes compute from each customer's own rate card rather than being generated, eliminating a class of AI failure where a hallucinated number costs real money.
  - Wrote a ten-rule specification governing the agent's writing voice and enforced it in the test suite**, failing the build on any violation.
  - Engineered for auditability and trust:** unreachable-model calls surface as distinct retryable states instead of silently degraded output, every outbound message writes a timestamped outbox row with a reason, and opt-outs are enforced at every send path.
  - Architected multi-tenant support scaling from a single owner-operator to multi-location businesses, with role-separated access and strict data isolation between locations.
  - Owned go-to-market end to end: positioning, the marketing site, and a free assessment tool that quantifies the hours a shop loses each week.

#### Freelance Web Developer

2023 – Present

- Shipped custom sites for a concert videographer, a sports photographer, a marketing agency, and a rugby academy — handling requirements, design, build, and delivery independently for each client.
- Built a React-based prequalification assessment for a marketing agency to screen client readiness before intake.

#### Target — Food Department Team Member

Jun 2024 – Present

- Customer service in a fast-paced retail environment; inventory management and department organization.

#### Coldwell Banker, Dan Firks Team — Real Estate Runner

Jun 2025 – Jun 2026

- Coordinated and staged open houses for one of the area's top-performing real estate teams.

### TECHNICAL SKILLS

|           |                                                                                                                               |
|-----------|-------------------------------------------------------------------------------------------------------------------------------|
| Languages | Python, JavaScript, HTML, CSS, React                                                                                          |
| AI        | Prompt engineering, LLM integration, AI agent design, human-in-the-loop approval workflows, automated testing of model output |
| Systems   | Multi-tenant architecture, role-based access, audit logging, messaging/SMS, payment processing, scheduling & invoicing        |

### EDUCATION

|                                                   |           |
|---------------------------------------------------|-----------|
| University of Illinois Urbana-Champaign           | Exp. 2030 |
| Gies College of Business — Finance & Data Science |           |
| Metea Valley High School                          | 2026      |
| GPA 4.25 — Aurora, IL                             |           |

### HONORS & ACTIVITIES

- Illinois State Scholar · Magna Cum Laude
- AP Scholar with Honor · Seal of Biliteracy
- National Honor Society — Member
- UNICEF Club — Board Member
- Vokel — grew a YouTube/Twitch channel to ~500 subscribers and Twitch Affiliate status